MA 21 - 2018
Spectroscopy II tutorial

Useful physical constants: h=6.626×10^{-34} J.s ; k=1.38×10^{-23} J.K^{-1} ; N_A=6.023×10^{23} mol^{-1}

Exercise 1: *Energy units*

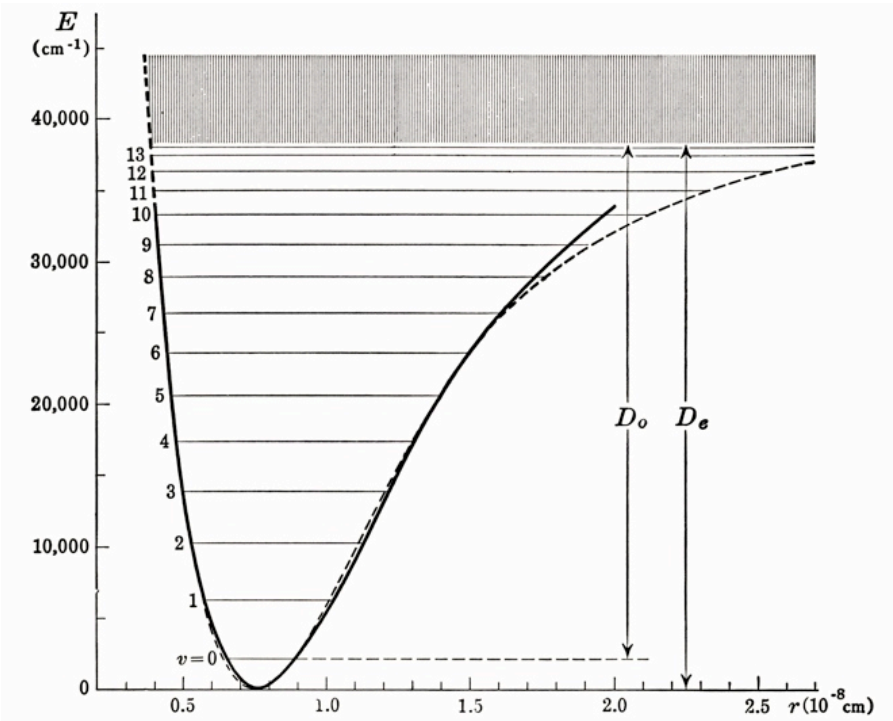
Convert the following spectroscopic quantities as indicated:

2000 cm^{-1}	to µm
0.15 nm	to Hz
500 nm	to cm^{-1}
9 GHz	to cm^{-1}

State in which region of the electromagnetic spectrum you would expect each corresponding transition to appear.

Exercise 2: *Bond Dissociation energy*

The dissociation energy D_0 corresponds to the energy required to break the chemical bond of a molecule. It is represented below in the case of H_2.



It is possible to derive the value of D_e (and then D_0) from the anharmonic oscillator model. In this model the molecular energy levels are given by:

$$E(v) = \omega_e (v + ½) - \omega_e \chi_e (v + ½)^2$$

Where v is the vibration quantum number, χ_e is the anharmonicity constant (such as χ_e<<1). In this formula, E and ω_e are expressed in cm^{-1}.

1) Calculate the energy difference ΔE between two consecutive levels.

2) What happens to ΔE when one becomes closer to the dissociation limit? Show that the highest possible value for the vibration quantum number is $v_{\max} = \frac{\omega_e}{2\omega_e \chi_e}$

3) Derive the Birge-Spooner formula: $D_e \simeq \frac{\omega_e^2}{4\omega_e \chi_e}$

4) In the case of H_2, ω_e = 4401.21 cm^{-1} and $\omega_e \chi_e$ = 121.33 cm^{-1}. Compute v_{max}, D_e et D_o. Express D_o in kcal/mol.

<u>Exercise 3:</u> *Thermal Population of Energy Levels*

1) Which one of the following molecules has the most populated vibrational level v=1 at 160°K : $^{35}Cl_2$, $^{35}Cl^{37}Cl$ *and* $^{37}Cl_2$?

2) What rotational level is the most populated for a nitrogen molecule ($B=2.01\ cm^{-1}$) at 500°K and at 100°K?

<u>Exercise 4:</u> *Rotational spectroscopy*

The rotational spectrum of $^{79}Br^{19}F$ shows a series of equidistant lines 0.71433 cm^{-1} apart. The atomic masses are $^{79}Br=131.03\times10^{-27}$ kg and $^{19}F=31.55\times10^{-27}$ kg.

1) Why are the lines equidistant?

2) Calculate the rotational constant B, and hence the moment of inertia and bond length of the molecule.

3) Determine the wavenumber of the J=9 → J=10 transition.

4) Calculate the number of revolutions per second which the BrF molecule undergoes when in (a) the J=0 state, (b) the J=1 state, and (c) the J=10 state (use $E_{rot}= ½\ I\omega^2$).

<u>Exercise 5:</u> *Vibrational spectroscopy (2013 exam)*

A diatomic oxide presents an IR absorption line at 1876 cm^{-1}. Moreover, it is known that the bond force constant (i.e. the bond stiffness) is k = 1550 $N.m^{-1}$.
Using these data, identify the molecule.

<u>Exercise 6:</u> *N_2 and H_2.*

Let's consider the following diatomic molecules:
N_2: molar mass M_N=14.007 g/mol, inter-atomic distance $r_e=1.094\times10^{-10}$ *m.*
H_2 : molar mass M_H=1.008 g/mol, inter-atomic distance $r_e=0.7417\times10^{-10}$ *m.*

1) Compute the energy of the first three rotational levels of N_2 (use the rigid rotor model).

2) a- Compute the energy (in cm^{-1}) of the first three lines of the rotational spectrum of H_2, in the rigid rotor model.

b- In the case of the harmonic oscillator, compute the force constant describing the bond between the two hydrogen atoms. The energy of the first vibrational transition (v=0 ⟶ v=1) is 4159.5 cm^{-1}.

c- What is the population ratio between the first two vibrational states at T=300K and T=1000K ?

d- Both hydrogen atoms are replaced by deuterium atoms (that is, hydrogen isotope whose nucleus is made by one proton + one neutron). Assuming that the vibrational force constant is not affected by this isotopic substitution, compute the new energy of the first vibrational transition. The molar mass of deuterium is M_D=2g/mol.

<u>Exercise 7:</u> *Spectrum of carbon monoxide*

In the *CO* molecule, the inter-atomic distance at equilibrium is $r_e=1.128\times10^{-10}$ *m*. The atomic masses of carbon and oxygen are respectively M_C=12.0 g/mol and M_O=15.995 g/mol.

1) Using the model of the rigid rotor, compute the energies (in cm^{-1}) of the first rotational levels of the CO molecule. Plot in an energy diagram the first five levels.

2) The force constant describing the C-O bond is $k_{C\text{-}O}$ = 1902 $N.m^{-1}$.

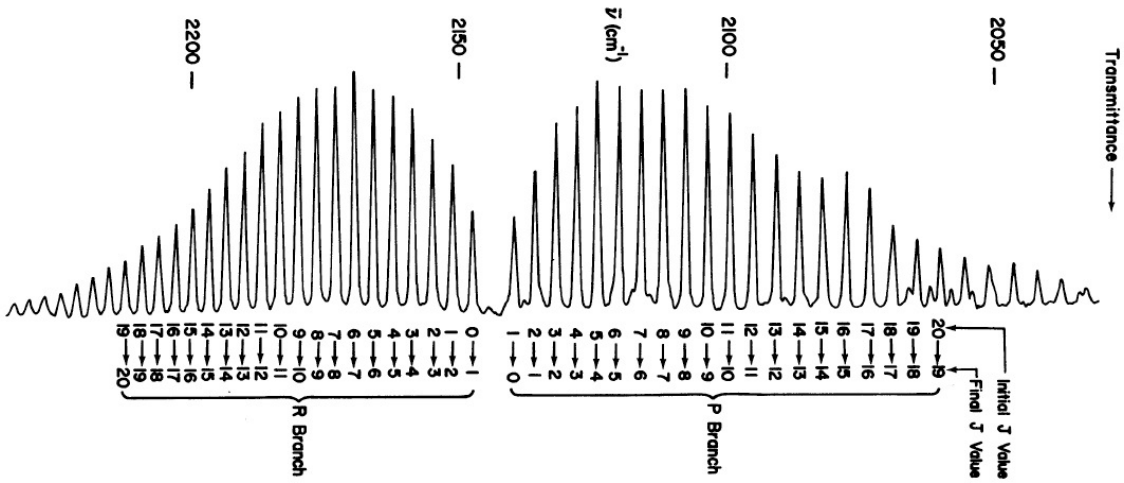
a- Compute the energy (in cm^{-1}) of the first two vibrational levels of CO, in the case of the harmonic approximation.

b- The experimental value of the first vibrational transition (v=0→v=1) is 2143 cm^{-1}, while the transition v=0→v=2 (first overtone) is at 4260 cm^{-1}. What kind of correction should be applied to the harmonic model? Compute the values of the corrective terms.

3) a- Compute the population ratio between the first two vibrational levels at T=300°K and T= 1000°K.

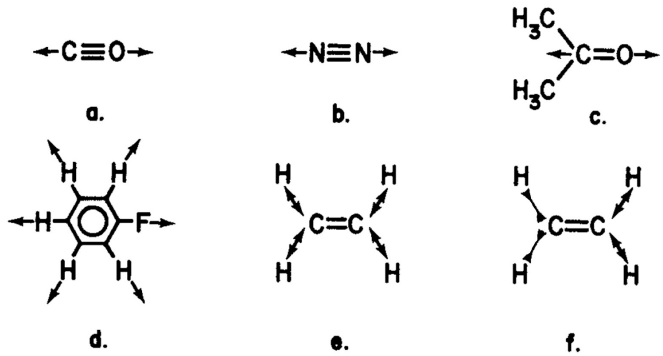
b- What rotational state is the most populated at T=300°K ?

4) Do we observe on the following spectrum the presence of the natural isotope of carbon ^{12}C, that is ^{13}C?



Exercise 7: *Selection rules*

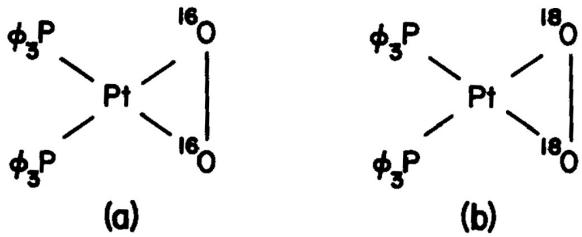
Are the molecular vibrations described below IR-active?



Exercise 8:

The $PtP_2{}^{16}O_2$ complex (a) exhibits an IR absorption band at 822 cm^{-1} attributed to the stretching mode of the O-O bond. We assume that during the movement of the oxygen atoms, the stretching of the Pt-O bond is negligible.

Assuming that the O-O stretching mode is described by an harmonic oscillator, compute the energy position of the absorption band of the isotopic compound $PtP_2{}^{18}O_2$ (b) ?



Exercise 9:

You need to analyze the Anti-Stokes part of the Raman spectrum of a given sample. How could you increase the intensity of this part of the spectrum?

Exercise 10:

In appendix, you will find the Raman and IR spectra of two aromatic compounds:
- benzene, C_6H_6.
- deuterated benzene, C_6D_6.

On each spectrum, the IR spectrum is on top and the Raman spectrum below.

Associate to each compound its corresponding spectrum. *Hint: compute the energy shift induced by the isotopic substitution H→D.*

Exercise 11: *Vibrational spectroscopy of ethylene (C_2H_4) and ethane (C_2H_6).*

The purpose of this exercise is to estimate the force constants of the C-C and C=C bonds (k_{C-C} and $k_{C=C}$).

1) How many normal modes of vibration are possible in ethane and ethylene?

2) Are the modes presented on figure 1 IR-active or Raman-active?

3) In the harmonic approximation, it is possible to define for each vibration a force constant *k.* Show that *k* can be written in the usual form:

$$k = \left(59.10^{-8}\right)\mu\,\tilde{\lambda}^2$$

where *k* is in N.cm^{-1}, μ is the reduced mass in g.mol^{-1} and $\tilde{\lambda}$ is the wavenumber (in cm^{-1}).

4) As a first approximation, it is possible to estimate $k_{C=C}$ et k_{C-C} assuming that the modes at 1623 cm^{-1} (for ethylene) and 993 cm^{-1} (for ethane) are mainly due to the stretching of the carbon-carbon bond. Compute these first values.

5) The exact values of $k_{C=C}$ et k_{C-C} are respectively 9,6 and 4.5 N.cm^{-1}. Why these values are different from the ones previously computed? In other words, explain the limits of the approximation made in question 4.

Definition of the Newton : 1 N = 1 kg.m.s^{-2}
Avogadro number: N_A=6.022.10^{23} mol^{-1}
Molar mass of carbon: 12g/mol ; hydrogen: 1g/mol

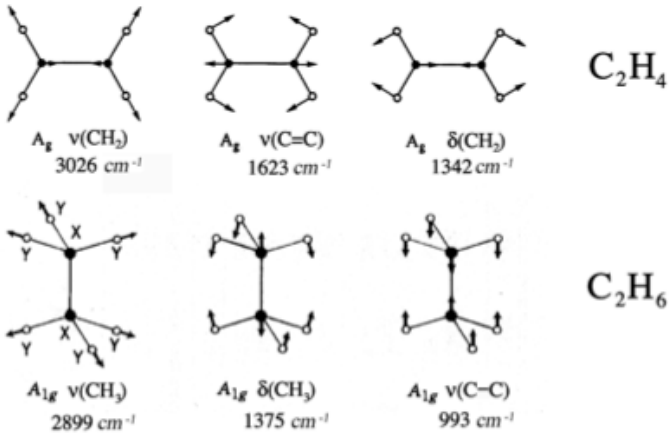


Figure 1 : sketch of the normal modes of vibration for C_2H_4 and C_2H_6 molecules. ν stands for stretching modes and δ for bending modes.

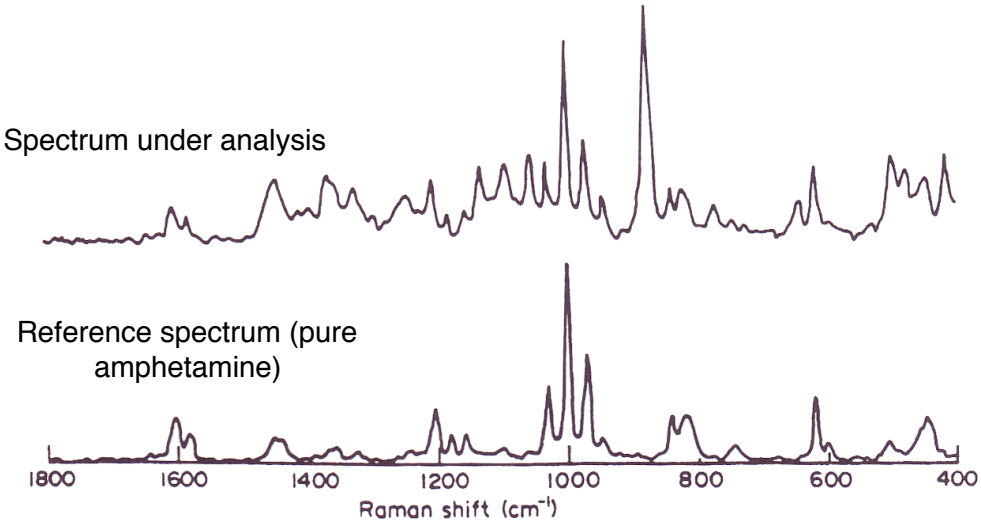
Exercise 12: *An unusual application of Raman spectroscopy (2009 exam)*

As a consequence of the global economic crisis and your difficulties to find a job, you are hired by Walter White, a notorious drug dealer operating in New Mexico. You are now Walter's assistant chemist, a nicely paid position… but where no mistake is allowed.

Walter asks you to analyze his new shipment of amphetamines, because he fears someones might have altered the product. You have at your disposition in the lab a Raman spectrometer, a reference sample of pure (uncut) amphetamine, and a sample of the suspicious product. The Raman spectra of both samples are presented below.

1) By considering these spectra, can you say if the drug has been altered? Carefully justify your answer, because Walter knows some serious chemistry.

2) If you think that the drug has been cut, propose an experimental procedure to identify the nature of the solvent in which the drug is mixed.



<u>Exercise 13:</u> *Spectroscopy of the HCl molecule (2010 exam)*

Assume the following data for the $^{1}H^{35}Cl$ molecule:

- bond length: 127.5 pm ;
- bond force constant: 516.3 $N.m^{-1}$;
- atomic masses: $m_H = 1.673\ 10^{-27}$ kg and $m_{Cl} = 58.066\ 10^{-27}$ kg

In the following, give all answers in cm^{-1}.

1) Calculate the energy of the first two vibrational levels.

2) Calculate the rotational constant B.

3) Sketch on an energy diagram the first three lines from P and R branches of the rovibrational spectrum of HCl. Those lines are noted $P_{(1)}$, $P_{(2)}$, $P_{(3)}$, $R_{(0)}$, R(1) and $R_{(2)}$.

4) Compute the energies (wavenumbers) associated to those six lines.

5) Is it possible to observe the Q branch on this molecule? Why?

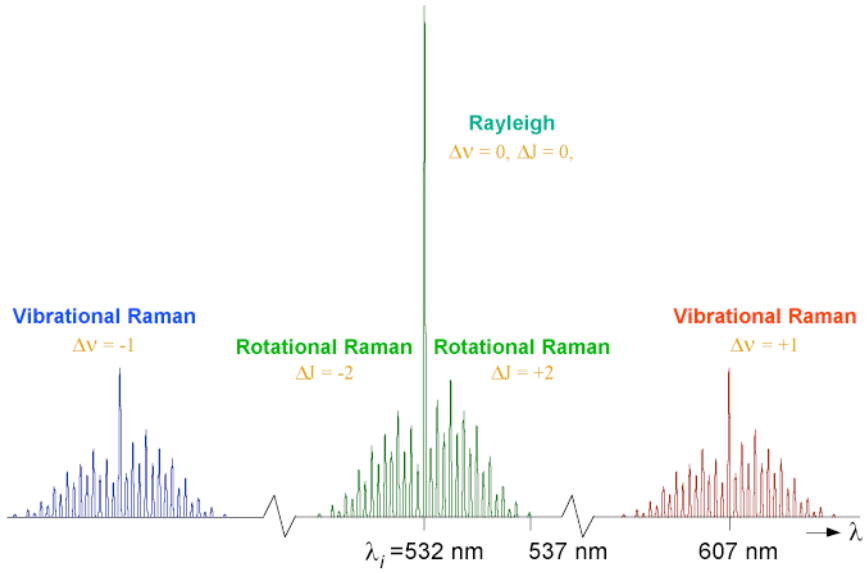
6) Sketch the expected rotation-vibration spectrum of HCl, including the *approximate* intensity distribution.

Exercise 14: *Raman Spectroscopy* (2011 exam)

Below is plotted the Raman spectrum of the N_2 molecule. It was recorded in gas phase with a monochromatic laser source emitting at λ_i = 532 nm. On the spectrum are also indicated the selection rules for each spectral line.

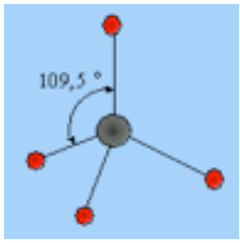
1) Starting from this spectrum, give in cm^{-1} the vibration frequency of N_2.

2) Comment on the relative intensities of the two vibrational lines. How the spectrum will look like in a liquid or solid medium?

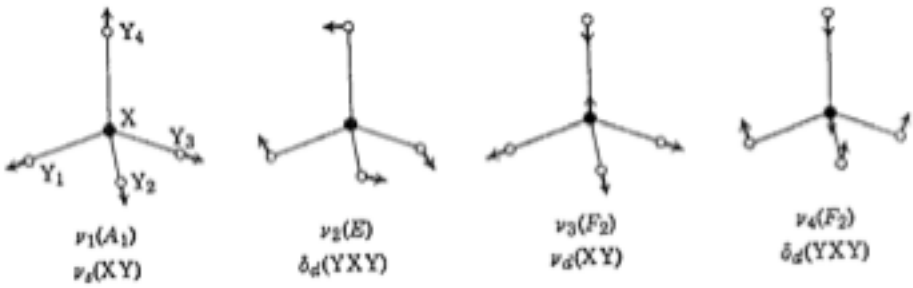


Exercise 15: *Infrared activity of methane* (2016 exam)

Methane (CH_4) is a tetrahedral molecule, as sketched here.



According to group theory, tetrahedral molecules with formula XY_4 exhibit four normal modes of vibration noted v_1, v_2, v_3, and v_4. The associated atomic motions are shown below. Note that for modes v_1 and v_2 the central atom X is not moving.

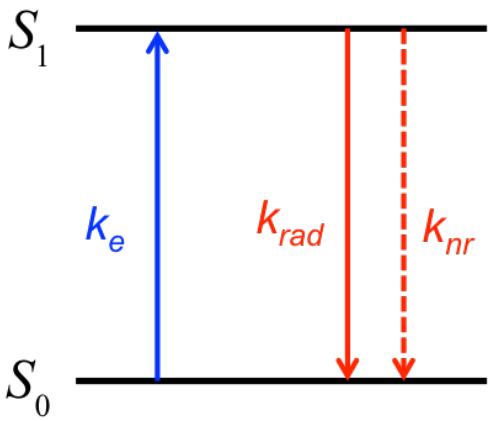


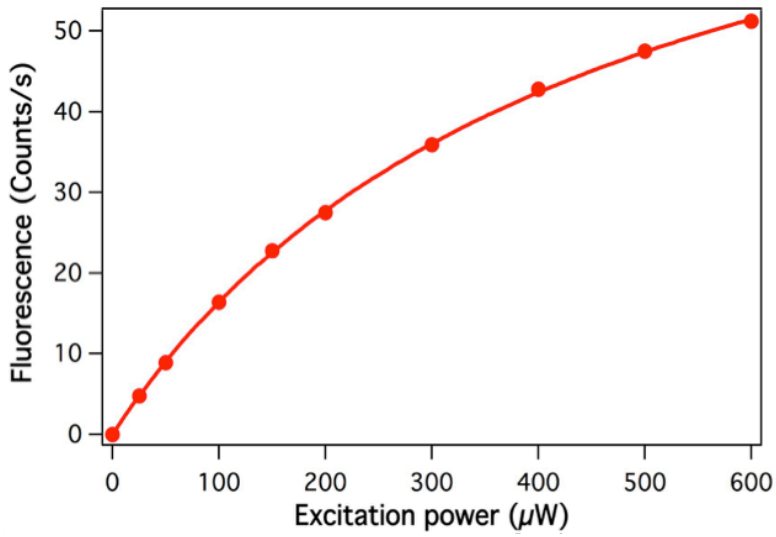
1) Are those modes active in infrared (IR) absorption spectroscopy?

2) It possible to demonstrate using group theory that modes v_3 and v_4 are three-fold degenerated, while the mode v_2 is two-fold degenerated. What is the degree of degeneracy of mode v_1?

Exercise 16: *Two-level description of fluorescent molecules*

We model a fluorescent molecule as a two-level quantum system (see the figure below, left panel), with two electronic levels S_0 and S_1. We denote k_{rad} and k_{nr} the radiative and non-radiative emission rates (from S_1 to S_0), and k_e the excitation rate (from S_0 to S_1).

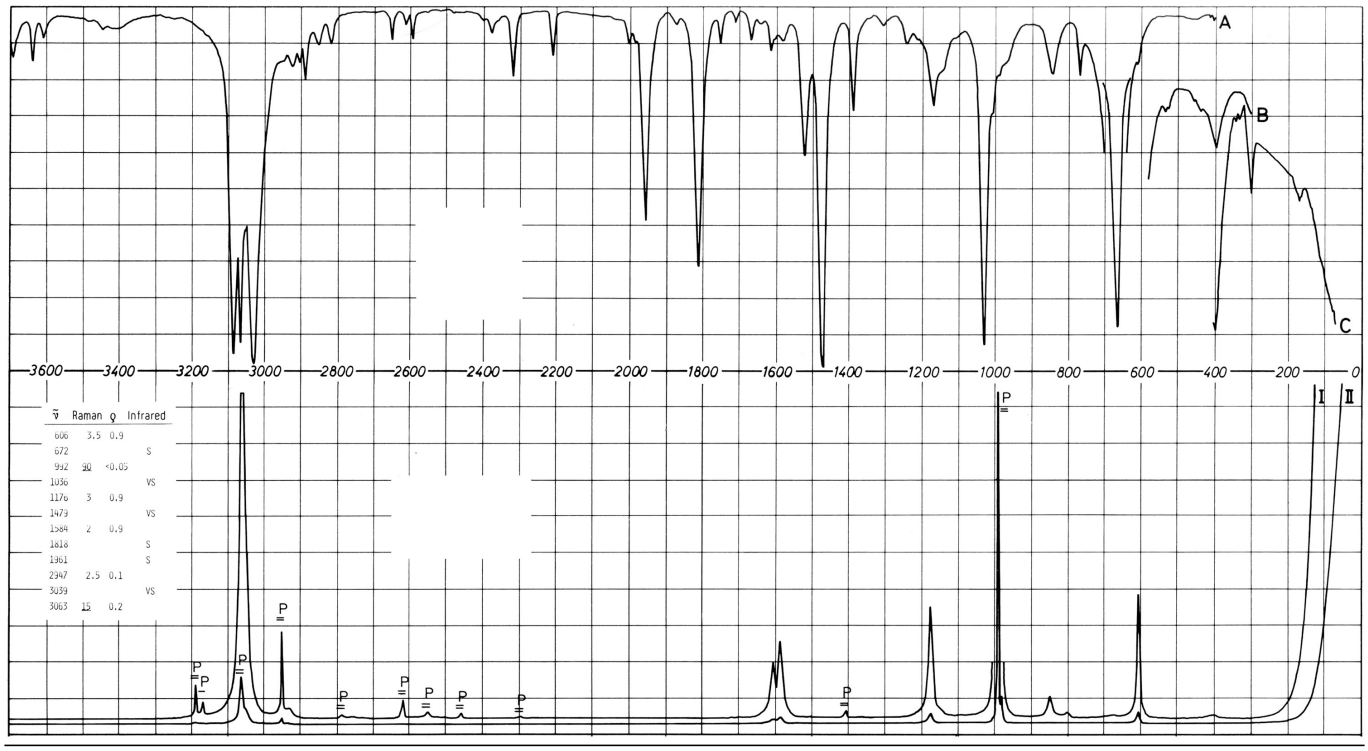
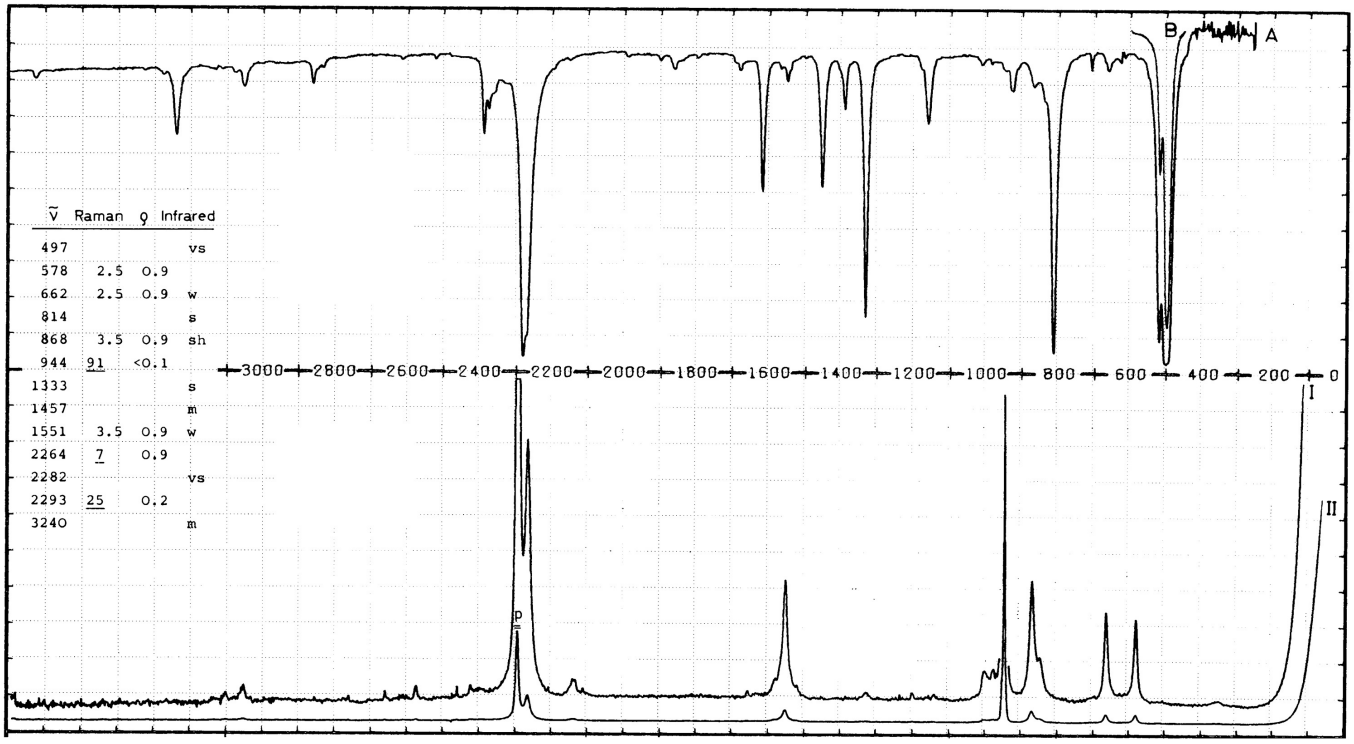




1) Write the excitation rate k_e as a function of the excitation cross-section σ_{abs} and the incident photon flux Φ.
2) We denote n_0 and n_1 the population of the levels S_0 and S_1, respectively, and N the total number of molecules. What are the variations in the populations, dn_0/dt and dn_1/dt?
3) Assuming steady state, compute the value of n_1 as a function of N, σ_{abs}, Φ and the total decay rate k_{tot}.
4) Compute the number of photons emitted per second W_{fluo} as a function of N, σ_{abs}, Φ, the quantum yield η and the saturation photon flux $\Phi_s = k_{tot}/\sigma_{abs}$.
5) The right panel of the figure shows an experimental measurement of the emitted fluorescence per molecule for Alexa-647 molecules diluted in water, as a function of the excitation power. Is this measurement compatible with the result obtained in the previous question?

Appendix:

Spectrum 1:



Spectrum 2: